

Project Submission Pro-Forma

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- ☒ MSc in Applied Artificial Intelligence
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- ☐ MSc in Cyber Security Management
- ☐ MSc in e-Business Management
- ☐ MSc in Engineering Business Management
- ☐ MSc in Games Engineering
- ☐ MSc in Healthcare Operational Management
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- ☐ MSc in International Trade, Strategy and Operations
- ☐ MSc in Management in Business Excellence
- ☐ MSc in Programme & Project Management
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- ☐ MSc in Sustainable Automotive Engineering

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- ☐ An abstract of the work completed
- ☐ A declaration of my contribution to the work and its suitability for the degree
- ☐ A table of contents
- ☐ A list of figures & tables (if applicable)
- ☐ A glossary of terms (where appropriate)
- ☐ A clear statement of my project objectives
- ☐ A full reference list (the Harvard referencing style is recommended for WMG)
- ☐ An appendix containing email confirmation of ethical approval or waiver

Ethical approval number or waiver status: _____

I understand that this dissertation will be stored for potential access by third parties e.g. staff/student training purposes, access by external examiners and professional statutory and regulatory bodies.

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Date: _____



**From Ingestion to Impact: Design and Evaluation
of an Evidence-First Multi-Agent AI Pipeline for
Auditable UK Early-Stage Portfolio Reporting and
Bounded Company Intelligence**

by

Oskar Rodziewicz

Dissertation submitted in partial fulfilment for the Degree of Master of Science in
Applied Artificial Intelligence

WMG
University of Warwick

Submitted _____

Declaration

I have read and understood the rules on cheating, plagiarism and appropriate referencing as outlined in my handbook and I declare that the work contained in this assignment is my own, unless otherwise acknowledged.

No substantial part of the work submitted here has also been submitted by me in other assessments for this or previous degree courses, and I acknowledge that if this has been done an appropriate reduction in the mark I might otherwise have received will be made.

Project definition for my degree (as copied from the Specific Course Regulations for the course)

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My project relates to this definition in the following way

Candidate to complete the one-page project-alignment declaration.

Abstract

Early-stage portfolio reporting combines heterogeneous submissions with incomplete and changing public evidence. This creates risks of entity mismatch, blank-versus-zero errors, temporally ineligible facts, hidden contradictions, and fluent but unsupported narrative. This dissertation investigates the extent to which a bounded multi-agent artificial-intelligence workflow can automate data ingestion, validation, and report generation while preserving traceability, reliability, and human authority. It defines deterministic, independent-verification, manual, and human-review comparisons rather than assuming that additional agents improve quality.

The implemented research artefact is a loopback-only Python/FastAPI/SQLite application. Typed contracts preserve source classification, reporting periods, missing states, immutable hashes, and exact claim-to-evidence provenance. A fixed workflow separates planning, identity resolution, collection, extraction, normalisation, independent verification, composition, and human review. An implemented but live-unrun Company 360 case study accepts a Companies House number as its required intake, holds legal identity for named review, and uses OpenAI web search only to discover candidate URLs. Local controls include connection-to-validated-public-IP pinning with hostname TLS verification, guarded capture, immutable snapshots, bounded public-text extraction, exact-span validation, deterministic deck composition, and named approval. Classification gates are tested to block restricted or internal material from external-model processing. The scope excludes investment recommendations and speculative valuation, and documented pattern controls reject tested forms; this does not demonstrate universal enforcement across all formulations.

Engineering evaluation used fourteen labelled fictional cases covering missingness, mixed types and currencies, identity ambiguity, stale or inaccessible evidence, contradictions, duplicate support, unsupported claims, and prompt injection. Across three deterministic repetitions, the independent-verifier condition matched all expected case outcomes, with precision, recall, and F1 of 1.00, provenance completeness of 1.00, and an unsupported-claim rate of 0.00. The no-verifier baseline produced precision of 0.455, F1 of 0.625, provenance completeness of 0.818, and an unsupported-claim rate of 0.545. These are reproducible synthetic mechanism measurements, not evidence of superiority on operational data.

As at 27 August 2026, manual and human-review evaluation is a no-go because ethics, data, and participant authority are unverified. If authorisation remains absent at protocol freeze, the manual comparison will be reported as unanswered, the primary research question as only partially answered, and synthetic evidence will not be substituted. The public company/model path also remains live-unrun; official live adapters, a frozen public benchmark, provider-retention and cost evidence, and production/accessibility validation are outstanding. The contribution is therefore an auditable artefact and evaluation protocol, limited to controlled technical feasibility rather than reduced real-world effort, improved investment decisions, exhaustive web coverage, or production readiness.

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Glossary and Acronyms

Term	Meaning
AI	Artificial intelligence.
API	Application programming interface.
C0–C3	Manual, deterministic baseline, multi-agent verification, and multi-agent plus human-review evaluation conditions.
D0–D2	Synthetic development, authorised pilot, and sealed final out-of-sample dataset tiers.
HITL	Human in the loop.
LLM	Large language model.
OOS	Out of sample.
RQ	Research question.
SSRF	Server-side request forgery.
URL	Uniform resource locator.

Introduction

- 1.1 Early-stage portfolio reporting and company intelligence
- 1.2 Problem definition and motivation
- 1.3 Aim and research questions
- 1.4 Research objectives
- 1.5 Scope, contribution, and exclusions
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5.3 D0 fixture and comparison design

5.4 D0 synthetic mechanism results

5.5 Company-research contract and adversarial results

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Ethics, Governance and Limitations

7.1 Data classification, privacy, and purpose limitation

7.2 External-model processing and retention

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7.4 Investment and public-person boundaries

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Conclusion and Future Work

8.1 Proportionate answers to the primary question and RQ1–RQ3

8.2 Demonstrated contribution

8.3 Sequenced future evidence

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